Unit 26: The Definite Integral — Full Solutions

Warm-up

1. $\int_1^4 f(x)\,dx = 10$, find $\int_4^1 f(x)\,dx$.

Reversing the limits flips the sign:

$$\int_4^1 f(x)\,dx = -\int_1^4 f(x)\,dx = -10$$

2. $\int_0^3 f(x)\,dx = 7$, find $\int_0^3 5f(x)\,dx$.

$$\int_0^3 5f(x)\,dx = 5\int_0^3 f(x)\,dx = 5(7) = 35$$

3. $\int_2^2 f(x)\,dx$

By definition, integrating over an interval of zero width gives 0, regardless of what f is:

$$\int_2^2 f(x)\,dx = 0$$

4. $\int_0^2 f(x)\,dx = 4$, $\int_2^5 f(x)\,dx = 9$, find $\int_0^5 f(x)\,dx$.

By the additivity property:

$$\int_0^5 f(x)\,dx = \int_0^2 f(x)\,dx + \int_2^5 f(x)\,dx = 4 + 9 = 13$$

5. $\int_1^3 f(x)\,dx = 6$, $\int_1^3 g(x)\,dx = 2$, find $\int_1^3 [f(x) - g(x)]\,dx$.

$$\int_1^3 [f(x) - g(x)]\,dx = \int_1^3 f(x)\,dx - \int_1^3 g(x)\,dx = 6 - 2 = 4$$

6. $\int_0^4 f(x)\,dx = 8$, $\int_0^4 g(x)\,dx = 3$, find $\int_0^4 [2f(x) + 3g(x)]\,dx$.

$$\int_0^4 [2f(x) + 3g(x)]\,dx = 2\int_0^4 f(x)\,dx + 3\int_0^4 g(x)\,dx = 2(8) + 3(3) = 16 + 9 = 25$$

Standard

7. $\int_0^5 f(x)\,dx = 12$, $\int_3^5 f(x)\,dx = 4$, find $\int_0^3 f(x)\,dx$.

By additivity: $\int_0^5 f(x)\,dx = \int_0^3 f(x)\,dx + \int_3^5 f(x)\,dx$.

$$12 = \int_0^3 f(x)\,dx + 4 \quad \Rightarrow \quad \int_0^3 f(x)\,dx = 8$$

8. $\int_1^6 f(x)\,dx = 20$, $\int_1^4 f(x)\,dx = 9$, find $\int_4^6 f(x)\,dx$.

By additivity: $\int_1^6 f(x)\,dx = \int_1^4 f(x)\,dx + \int_4^6 f(x)\,dx$.

$$20 = 9 + \int_4^6 f(x)\,dx \quad \Rightarrow \quad \int_4^6 f(x)\,dx = 11$$

9. $\int_2^7 f(x)\,dx = 15$, find $\int_7^2 f(x)\,dx$ and $\int_7^2 [-3f(x)]\,dx$.

$$\int_7^2 f(x)\,dx = -\int_2^7 f(x)\,dx = -15$$

$$\int_7^2 [-3f(x)]\,dx = -3\int_7^2 f(x)\,dx = -3(-15) = 45$$

10. $g(x) = f(x) + 2$.

$$\int_1^5 g(x)\,dx = \int_1^5 [f(x)+2]\,dx = \int_1^5 f(x)\,dx + \int_1^5 2\,dx = \int_1^5 f(x)\,dx + 2(5-1) = \int_1^5 f(x)\,dx + 8$$

Answer: $\int_1^5 g(x)\,dx$ is exactly 8 more than $\int_1^5 f(x)\,dx$. This makes sense geometrically: shifting the entire graph up by 2 units adds a rectangular strip of height 2 and width 4 (the length of the interval) underneath the new curve, and $2 \times 4 = 8$.

11. f continuous, $f(x) \geq 0$ on $[-3,3]$, $\int_{-3}^3 f(x)\,dx = 0$.

Conclusion: $f(x) = 0$ for every x in $[-3,3]$.

Reasoning: suppose, for contradiction, that f were positive at some point in the interval. Since f is continuous, it would then have to stay positive on some small stretch of x-values surrounding that point (a continuous function can't jump instantly back to zero). That small positive stretch would contribute a strictly positive amount of area to the integral. But we're told the total integral is exactly 0, and since $f(x) \geq 0$ everywhere else too (no negative area to cancel it out), there's no way to make the total come out to 0 if any part of it is strictly positive. So f must be 0 at every single point in the interval.

12. $m = 2$, $M = 6$ on $[1,4]$ (interval length $= 3$).

$$m(b-a) \leq \int_1^4 f(x)\,dx \leq M(b-a)$$

$$2(3) \leq \int_1^4 f(x)\,dx \leq 6(3)$$

$$6 \leq \int_1^4 f(x)\,dx \leq 18$$

Challenge

13. Show $1 \leq \int_0^1 \sqrt{x^2+1}\,dx \leq \sqrt{2}$.

Let $f(x) = \sqrt{x^2+1}$. Since x^2+1 is increasing for $x \geq 0$ (as x grows, x^2 grows), and the square root function preserves increasing order, $f(x)$ is also increasing on $[0,1]$.

So the minimum of f on $[0,1]$ occurs at $x = 0$: $f(0) = \sqrt{0+1} = 1$.

The maximum occurs at $x = 1$: $f(1) = \sqrt{1+1} = \sqrt{2}$.

By the Max-Min Inequality, with $b-a = 1-0 = 1$:

$$1 \times (1) \leq \int_0^1 \sqrt{x^2+1}\,dx \leq \sqrt{2} \times (1)$$

$$1 \leq \int_0^1 \sqrt{x^2+1}\,dx \leq \sqrt{2}$$

This confirms the claim.

14. Bounds for $\int_0^2 (x^2+3)\,dx$.

$f(x) = x^2+3$ is increasing on $[0,2]$ (since $f'(x) = 2x \geq 0$ there). So the minimum is $f(0) = 3$, and the maximum is $f(2) = 4+3 = 7$.

With $b-a = 2$:

$$3(2) \leq \int_0^2 (x^2+3)\,dx \leq 7(2)$$

$$6 \leq \int_0^2 (x^2+3)\,dx \leq 14$$

15. Is $\int_{-2}^{2} (x^3-4x)\,dx = 0$?

Let $f(x) = x^3 - 4x$. Check whether it's odd:

$$f(-x) = (-x)^3 - 4(-x) = -x^3 + 4x = -(x^3-4x) = -f(x)$$

Since $f(-x) = -f(x)$, f is an odd function.

The interval $[-2, 2]$ is symmetric about 0. By the odd-function symmetry property:

$$\int_{-2}^{2} (x^3-4x)\,dx = 0$$

Yes — the integral equals 0, confirmed without any direct evaluation.

16. Can $\int_{-\pi}^{\pi} x^2 \cos x\,dx$ be simplified using symmetry?

Let $f(x) = x^2 \cos x$. Check its symmetry:

$$f(-x) = (-x)^2 \cos(-x) = x^2 \cos x = f(x)$$

(using the fact that $(-x)^2 = x^2$ is even, and $\cos(-x) = \cos x$ is even, so their product is even too.)

Since $f(-x) = f(x)$, f is an even function.

This integral does not vanish to zero — the even-function property only guarantees a simplification, not automatic cancellation (cancellation to zero only happens for odd functions). Instead, it simplifies to:

$$\int_{-\pi}^{\pi} x^2 \cos x\,dx = 2\int_0^{\pi} x^2 \cos x\,dx$$

This cuts the required computation in half — you'd only need to evaluate the integral over $[0, \pi]$ and double the result — but it doesn't eliminate the integral the way an odd function would.

17. Why does $\int_{-1}^{1} \left(x^{2024} \sin x + \frac{x^7}{1+x^2} \right) dx = 0$?

Classify the first term, $x^{2024} \sin x$: x^{2024} has an even exponent, so it's an even function. $\sin x$ is a well-known odd function. By the combination rule, even $\times$ odd $=$ odd. So the first term is odd.

Classify the second term, $\frac{x^7}{1+x^2}$: x^7 has an odd exponent, so it's odd. The denominator $1+x^2$ is a constant plus an even power, so it's even. By the combination rule, odd $\div$ even $=$ odd. So the second term is odd too.

Combine: odd + odd = odd. So the entire integrand is an odd function.

Since the interval $[-1,1]$ is symmetric about 0, and the whole integrand is odd, the odd-function symmetry property applies directly:

$$\int_{-1}^{1} \left(x^{2024} \sin x + \frac{x^7}{1+x^2} \right) dx = 0$$

No antiderivative techniques were needed at all — the symmetry argument alone fully settles the answer.

18. Is $\int_0^6 f(x)\,dx = 20$ consistent with $2 \le f(x) \le 5$ on $[0,6]$?

Using the Max-Min Inequality with $m = 2$, $M = 5$, and $b - a = 6$:

$$2(6) \le \int_0^6 f(x)\,dx \le 5(6)$$

$$12 \le \int_0^6 f(x)\,dx \le 30$$

Since $12 \le 20 \le 30$, yes, this is fully consistent — the given value of 20 falls comfortably within the guaranteed bounds.

Applied

19. $-50 \le p(t) \le 200$ on $[0,30]$ (interval length = 30).

$$-50(30) \le \int_0^{30} p(t)\,dt \le 200(30)$$

$$-1500 \le \int_0^{30} p(t)\,dt \le 6000$$

Answer: total monthly profit is between $-\$1500$ and \$6000.

20. $d(t)$ is odd on $[-6,6]$.

By the odd-function symmetry property:

$$\int_{-6}^{6} d(t)\,dt = 0$$

Interpretation: the net temperature deviation over the entire day sums to exactly zero — the amount by which mornings run below the daily average exactly balances the amount by which evenings run above it, so there's no net bias across the full day.

21. $\int_0^{10} r(t)\,dt = 150$, $\int_0^{10} s(t)\,dt = 90$.

$$\int_0^{10} [r(t)+s(t)]\,dt = \int_0^{10} r(t)\,dt + \int_0^{10} s(t)\,dt = 150+90 = 240$$

Answer: total combined output $= 240$ gallons.

22. $40 \le v(t) \le 65$ on $[0,3]$ (interval length $= 3$).

$$40(3) \le \int_0^3 v(t)\,dt \le 65(3)$$

$$120 \le \int_0^3 v(t)\,dt \le 195$$

Answer: total distance traveled is between 120 and 195 miles.